

Rule-Based FAQ Chatbot Using Python

Assignment Report

Submitted By

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Assignment Title: Rule-Based FAQ Chatbot in Python

Abstract

Chatbots are software applications that interact with users through text or voice-based conversations. They are widely used in customer support, education, healthcare, and e-commerce. This project presents the development of a Rule-Based FAQ Chatbot using Python. The chatbot is designed to answer frequently asked questions related to a College Help Desk. It uses Natural Language Processing (NLP) techniques such as text preprocessing and Regular Expression (Regex) pattern matching to identify user queries and provide appropriate responses.

The chatbot operates through a command-line interface and can answer multiple frequently asked questions regarding admissions, fees, courses, hostel facilities, library timings, attendance requirements, and other college-related information.

1. Introduction

With the increasing demand for automated customer support systems, chatbots have become an important application of Artificial Intelligence and Natural Language Processing. A chatbot enables users to obtain information quickly without requiring human intervention.

This project focuses on developing a simple rule-based chatbot that responds to user queries using predefined rules and regular expression matching. Unlike AI-based chatbots, this chatbot does not learn from data but relies on manually defined patterns and responses.

2. Problem Statement

Students and visitors frequently ask similar questions regarding admissions, courses, fees, hostel facilities, and examination details. Handling these repetitive queries manually consumes time and resources.

The objective of this project is to develop a College Help Desk Chatbot that can:

- Answer frequently asked questions.
 - Interact with users through the terminal.
 - Handle different ways of asking the same question.
 - Provide appropriate responses using pattern matching.
 - Handle unknown queries gracefully.
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3. Objectives

The main objectives of this project are:

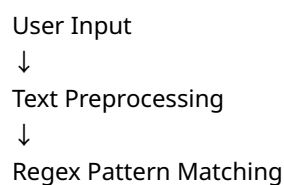
1. Understand the fundamentals of Natural Language Processing.
 2. Implement pattern matching using Regular Expressions.
 3. Design an interactive conversation flow.
 4. Process and clean user input.
 5. Develop a professional Python project structure.
 6. Build a chatbot capable of answering frequently asked questions.
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4. Technologies Used

Technology	Purpose
Python 3	Programming Language
VS Code	Development Environment
Regex (re module)	Pattern Matching
GitHub	Version Control
Terminal/Console	User Interaction

5. System Architecture

The chatbot follows the architecture shown below:



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Response Retrieval

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Display Response

Description

1. User enters a query.
 2. Input text is cleaned and converted to lowercase.
 3. Regex patterns are matched against predefined rules.
 4. If a match is found, the corresponding response is retrieved.
 5. If no match exists, a fallback response is displayed.
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6. Methodology

Step 1: Input Collection

The chatbot continuously accepts user input through the terminal using Python's `input()` function.

Step 2: Text Preprocessing

The input text undergoes preprocessing:

- Convert to lowercase.
- Remove special characters.
- Remove extra spaces.

Example:

Input:

"What are the College Timings?"

Processed Output:

"what are the college timings"

Step 3: Regex Pattern Matching

The chatbot uses Python's `re` module to compare user input against predefined regex patterns.

Example:

Pattern:

(hostel|accommodation|stay)

Queries matched:

- Is hostel available?
- Do you provide accommodation?
- Can students stay in hostel?

Step 4: Response Generation

If a pattern is matched, the corresponding answer is returned.

Example:

User:

"What is the fee structure?"

Bot:

"Fee structure varies by course. Please contact the accounts department."

Step 5: Fallback Response

If no pattern is matched:

Bot:

"Sorry, I couldn't understand your question."

7. Features Implemented

- Interactive terminal chatbot.
- Regex-based question matching.
- Input preprocessing.
- Support for multiple question formats.
- Fallback response for unknown queries.
- Easy-to-update response database.
- Modular code structure.

8. Frequently Asked Questions Covered

The chatbot can answer questions regarding:

1. College timings
2. Admissions
3. Required documents

4. Available courses
 5. Fee structure
 6. Hostel facilities
 7. Library timings
 8. Contact information
 9. Scholarships
 10. Class commencement dates
 11. Attendance requirements
 12. Results
 13. Transportation
 14. Examination Cell
 15. College location
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9. Sample Execution

Example 1

User:

What are the college timings?

Bot:

College timings are 9:00 AM to 5:00 PM.

Example 2

User:

Do you provide accommodation?

Bot:

Yes, hostel facilities are available for both boys and girls.

Example 3

User:

What is the tuition fee?

Bot:

Fee structure varies by course.

Example 4

User:

Hello

Bot:

Sorry, I couldn't understand your question.

10. Advantages

- Simple and easy to implement.
 - Fast response time.
 - Low computational requirements.
 - Easy maintenance.
 - Suitable for FAQ systems.
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11. Limitations

- Cannot learn from new data.
 - Responses are predefined.
 - Limited understanding of complex sentences.
 - Cannot maintain long conversations.
 - Performance depends on defined patterns.
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12. Future Enhancements

The chatbot can be enhanced by:

1. Integrating a database for storing FAQs.
 2. Adding speech-to-text functionality.
 3. Integrating Large Language Models (LLMs).
 4. Deploying as a web application using Flask.
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13. Conclusion

The Rule-Based FAQ Chatbot successfully demonstrates the use of Python, Natural Language Processing fundamentals, and Regular Expressions for building an interactive FAQ system. The chatbot effectively answers frequently asked college-related queries and provides a user-friendly interface through the terminal. This project serves as a foundation for understanding chatbot development and can be extended into more advanced AI-powered conversational systems in the future.